Phases of the half-filled extended Fermi-Hubbard model

Miquel Limón Vallés
Tutor: Bruno Juliá Díaz, Advisor: Artur Garcia Saez

Universitat de Barcelona - Barcelona Supercomputing Center

25th of June, 2025

Outline

1 Introduction

2 The Hubbard model

3 Tensor networks

4 Ground-state convergence

5 Results

6 Conclusions

Motivation

Band theory stems from a mean-field approximation for how electrons interact with a crystalline potential.

- De Boer & Verwey (1937): certain crystals with partially-filled bands display a lower conductivity than predicted by band theory (MnO, CoO, NiO).
- Mott & Peierls (1937): electron-electron repulsion might play a part in low electronic velocity.

$\rightarrow$ Different approach: consider particle interactions explicitly.

Goals

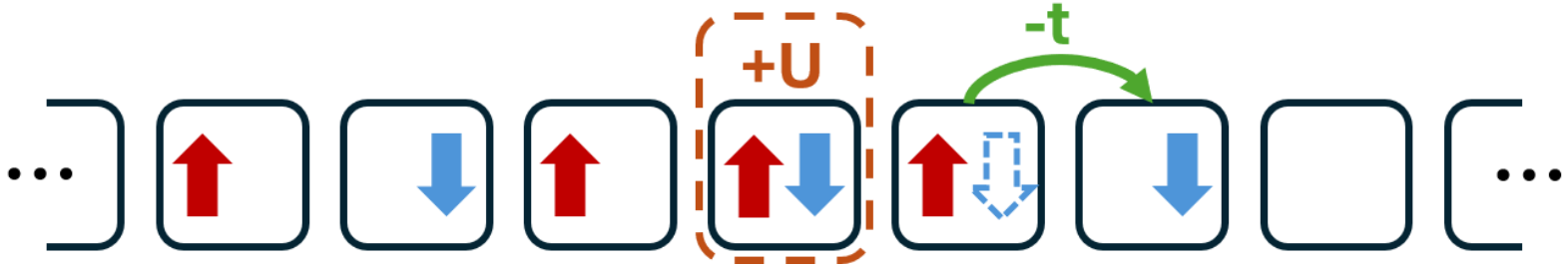
- Model electrons in a (1D) lattice as a quantum many-body problem.
- See if direct particle repulsion leads to a lower conductivity.

The Hubbard model

Simplest Hamiltonian for particle interaction in a lattice, with kinetic and on-site interaction terms.

$$\mathcal{H} = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i) + U \sum_i n_{\uparrow,i} n_{\downarrow,i}$$

Fock basis: $\{0, \uparrow, \downarrow, \uparrow\downarrow\} \rightarrow$ local Hilbert space of $d = 4$.

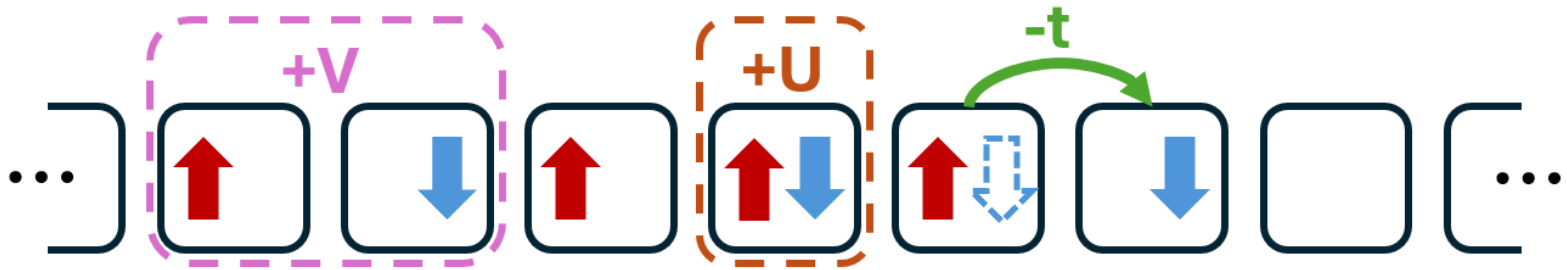


I'll be working with a lattice at half-filling: $N_{\text{particles}} = N_{\text{sites}}$.

The extended Hubbard model

Incorporates next-neighbour repulsion.

$$\mathcal{H}_{\text{extended}} = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i) + U \sum_i n_{\uparrow,i} n_{\downarrow,i} + V \sum_{\langle i,j \rangle} n_i n_j$$



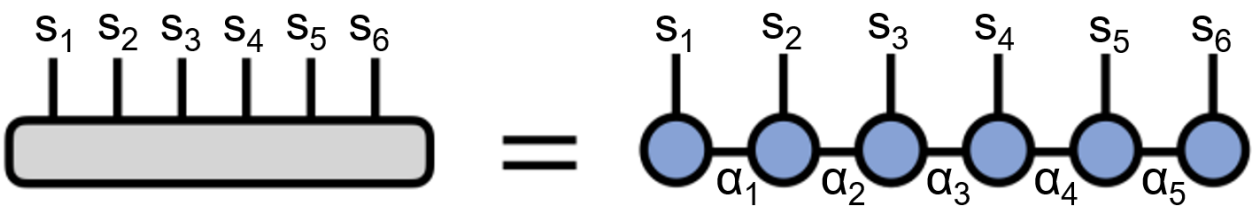
Wavefunctions and MPS

A wavefunction is a big matrix of probability amplitudes $\rightarrow$ tensor.

Matrix product state form:

$$|\Psi\rangle = \sum_{\{s_i\}} A^{s_1}_{\alpha_1} A^{s_2}_{\alpha_1\alpha_2} A^{s_3}_{\alpha_2\alpha_3} \ldots A^{s_N}_{\alpha_{N-1}\alpha_N} |s_1 s_2 \ldots s_N\rangle.$$

MPS are a lower-rank approximation of the original tensor.



Why tensor networks?

- Monte Carlo methods cannot fully explore the Hilbert space in fermionic simulations (sign problem), tensor network-based methods can.
- 1D gapped systems obey an area law for entanglement entropy.

$\rightarrow$ We can correctly obtain ground states with a limited χ!

Imaginary time evolution

A random wavefunction Ψ is a superposition of Hamiltonian eigenstates:

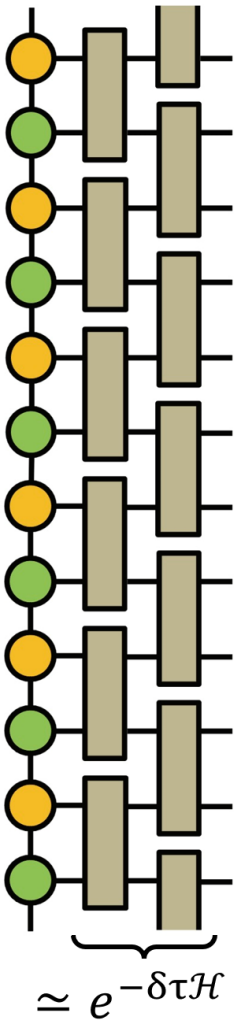
$$\Psi(x,t) = \sum c_n \psi_n(x) e^{-iE_n t/\hbar}$$

$$\downarrow \tau = it$$

$$\Psi(x,\tau) = \sum c_n \psi_n(x) e^{-E_n \tau/\hbar}$$

Ground-state can be achieved by applying small τ steps, $e^{-\delta\tau\mathcal{H}}$.

Time-evolving block decimation

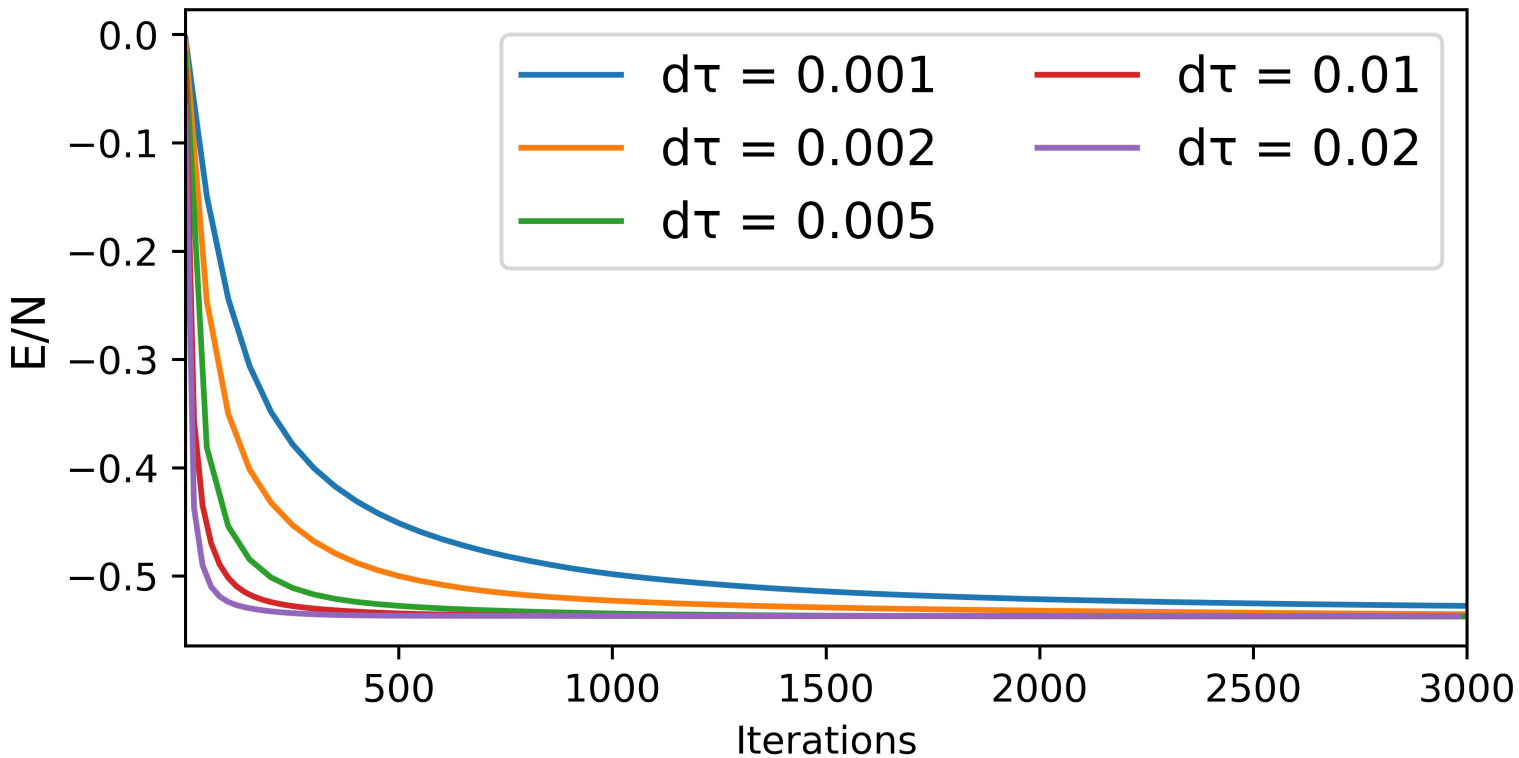


Local hamiltonians can be split into two-site gates that evolve the system locally. For a $e^{-\delta\tau\mathcal{H}}$ step and a 1D system of size N:

- Without TEBD: $O(d^{2N})$
- With TEBD: $O(N\chi^3 d^3)$

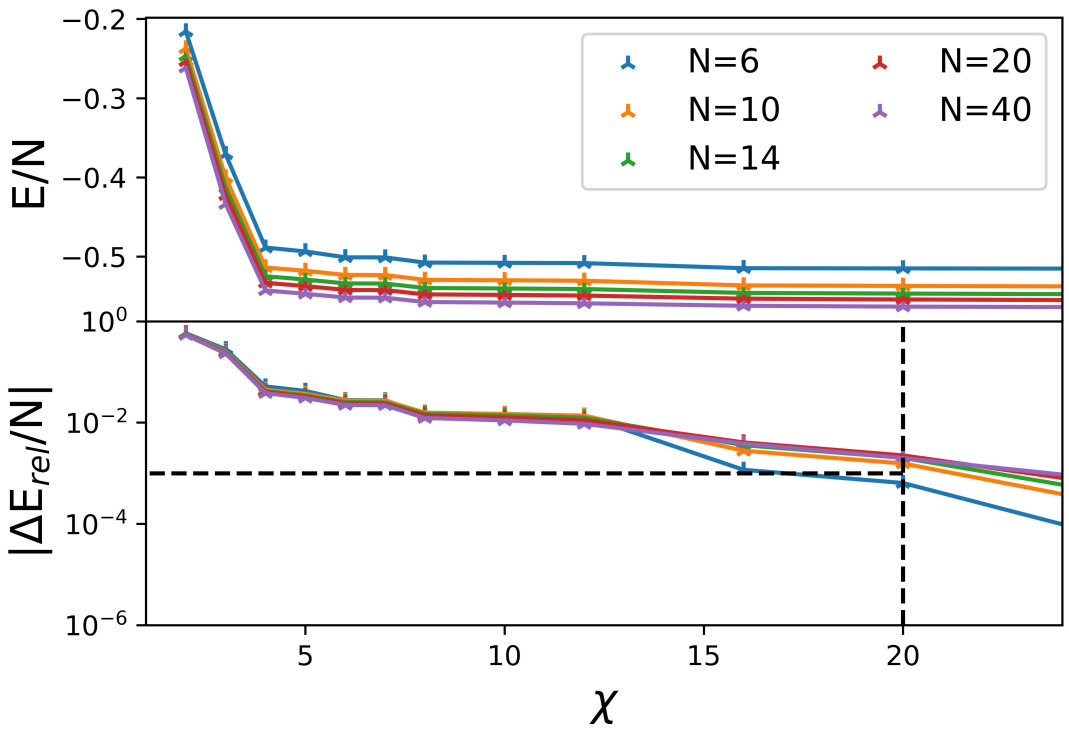
$\rightarrow$ TEBD scales linearly with system size!

Convergence & area law



Energy converges exponentially to a certain value. (N=10, U=4)

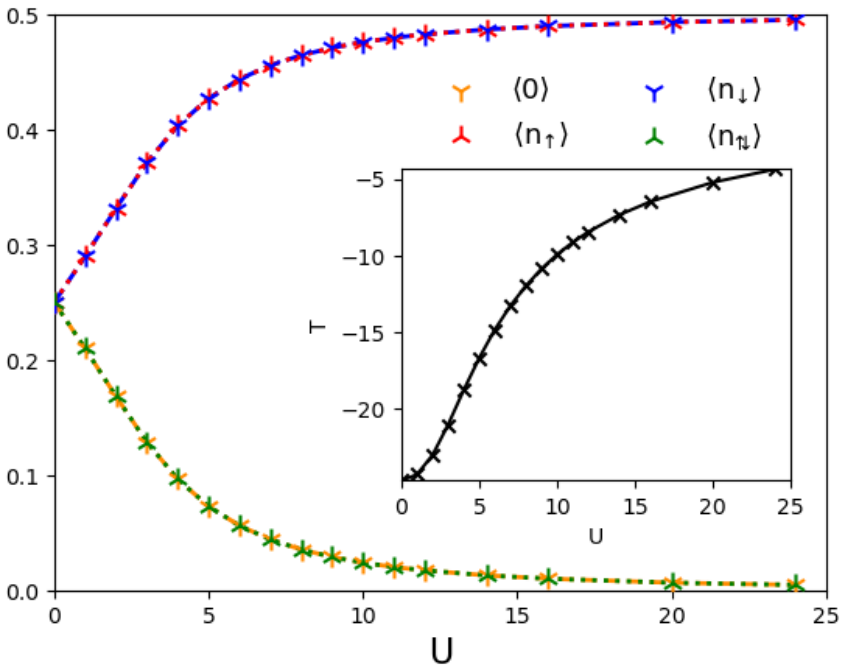
Convergence & area law



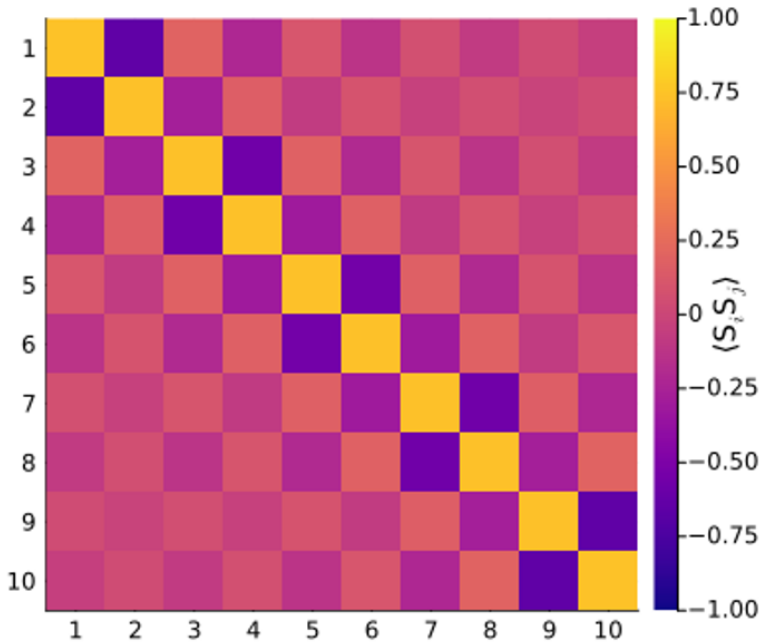
Relative energy error reaches $\sim 0.1\%$ for $\chi = 20$. (U=4)

From now on, $\chi = 20$.

On-site repulsion (regular model)



Probability of expected site states & kinetic energy T. (N=20)

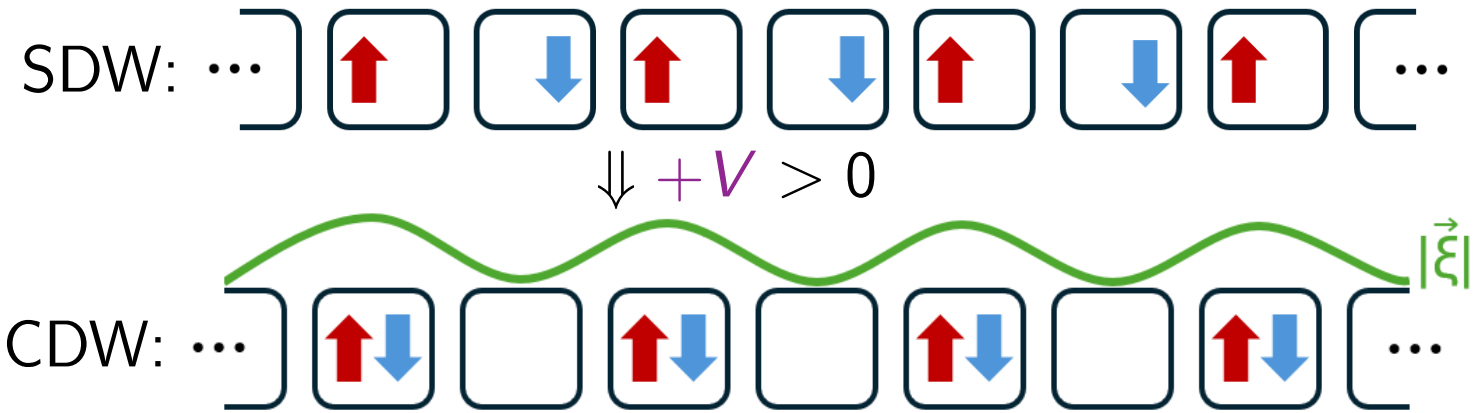


Spin-spin correlation matrix. (N=10, U=20)

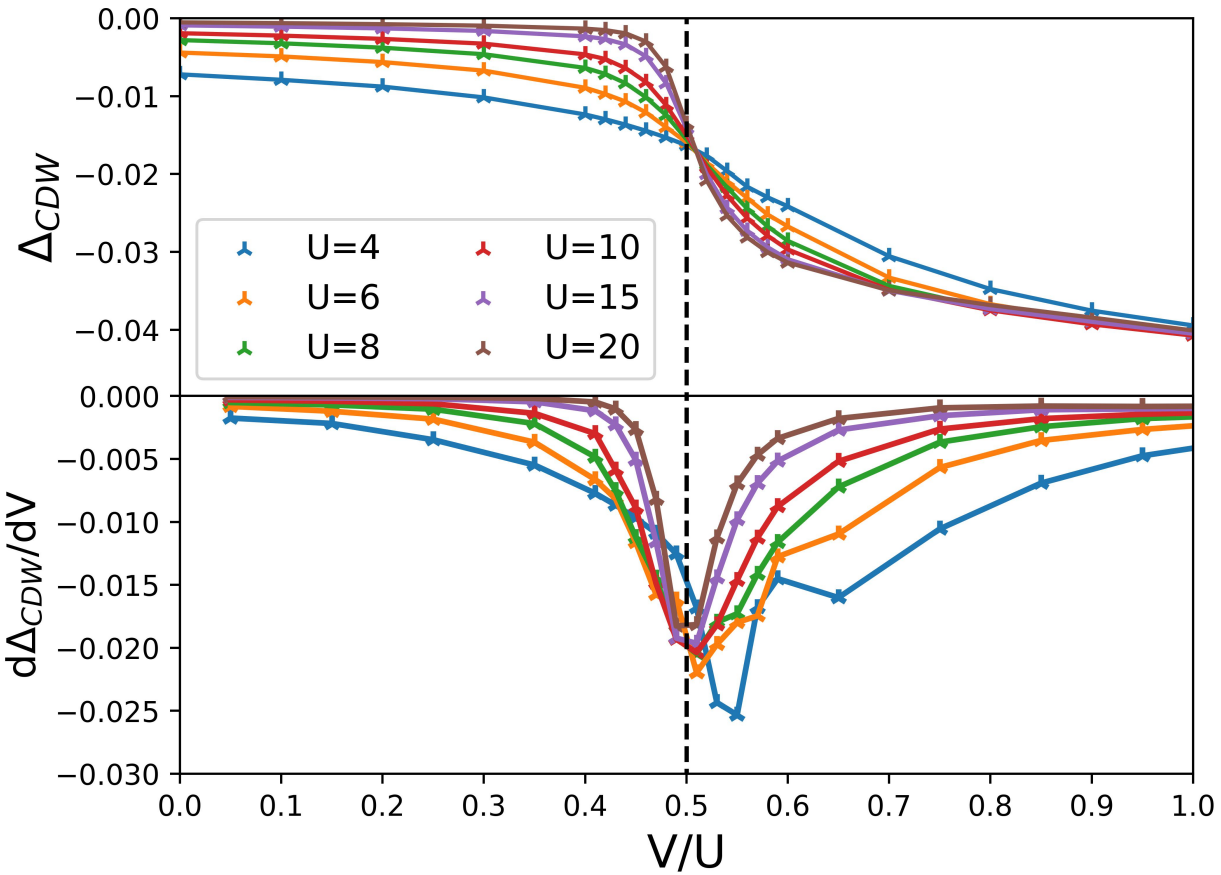
... + next-neighbour repulsion (extended model)

Transition from spin-density wave to charge-density wave.

$$\text{Order parameter: } \Delta_{CDW} = \frac{1}{N}\sum_i (-1)^i \langle n_{i,\uparrow\downarrow}\rangle$$



... + next-neighbour repulsion (extended model)



A phase transition at $V = U/2$ can be seen. (N=20)

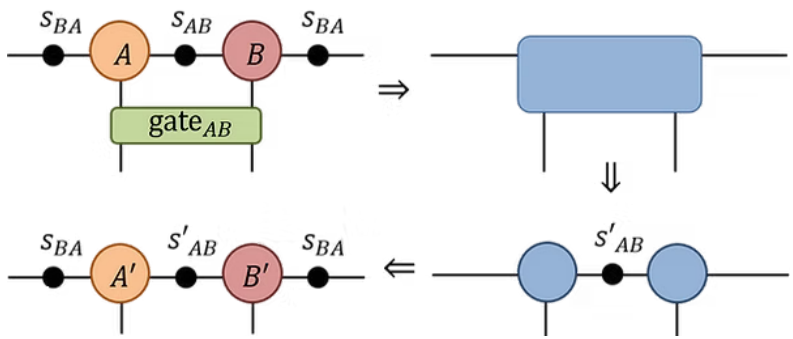
Conclusions

- Particle repulsion in a lattice gives rise to additional insulating phases at half-filling.
- Strong on-site repulsion leads to antiferromagnetic fermion ordering.
- Tensor networks allow for great optimisation of quantum simulations.

Simple update

This process takes place every time a two-site gate is applied onto a 1D MPS:

1. Site fusion
2. Gate application
3. SVD to recover MPS form



Singular Value Decomposition (SVD)

SVD is an algebraic operation for splitting a matrix A of dimension $m \times n$ into a rotation V of dimension $n \times n$, a scaling diagonal matrix S of dimension $m \times n$, and another rotation U of dimension $m \times m$:

$$A = USV^*$$

The eigenvalues in S are ordered by decreasing magnitude, which allows for an easy implementation of truncation and thus bond dimension limiting.

Superexchange

Superexchange is a second-order perturbation: although a particle is localised in a site, it can virtually hop to an adjacent site to increase the amount of movement in the system... in some cases. Consider an $\uparrow$ particle:

- $\downarrow$ neighbour: energetic cost of U.
- $\uparrow$ neighbour: Pauli exclusion principle.

$\rightarrow$ Particles will favour opposite-spin neighbours.

